MSG No. 101.185 P_C4_2cm [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `2.mm`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4,7,8]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[2,3,5,6]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[9,12,15,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,11,13,14]`

Table 2: Wyckoff site: `4b`, site symmetry: `2.m'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4,15,16]`
2	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[2,3,13,14]`
3	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,6,10,11]`
4	$\left[\frac{1}{2},\ 0,\ z\right]$	`[7,8,9,12]`

Table 3: Wyckoff site: `4c`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,7,12,16]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[2,6,11,13]`
3	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[3,5,10,14]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[4,8,9,15]`

Table 4: Wyckoff site: `8d`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,8]`
2	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	`[2,5]`
3	$\left[x,\ -x,\ z+\frac{1}{2}\right]$	`[3,6]`
4	$[-x,\ -x,\ z]$	`[4,7]`
5	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	`[9,16]`
6	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,13]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[11,14]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z\right]$	`[12,15]`

Table 5: Wyckoff site: `8e`, site symmetry: `..m'`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	`[1,16]`
2	$\left[\frac{1}{2}-x,\ x,\ z+\frac{1}{2}\right]$	`[2,13]`
3	$\left[x+\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	`[3,14]`
4	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	`[4,15]`

continued ...

Table 5

No.	position	mapping
5	$\left[-x,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[5,10]
6	$\left[x,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[6,11]
7	$\left[\frac{1}{2}-x,\, -x,\, z\right]$	[7,12]
8	$\left[x+\frac{1}{2},\, x,\, z\right]$	[8,9]

Table 6: Wyckoff site: 16f, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$\left[-y,\, x,\, z+\frac{1}{2}\right]$	[2]
3	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	[3]
4	$[-x,\, -y,\, z]$	[4]
5	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[5]
6	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[6]
7	$[-y,\, -x,\, z]$	[7]
8	$[y,\, x,\, z]$	[8]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[9]
10	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[10]
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[11]
12	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[12]
13	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[13]
14	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[15]
16	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[16]